

SLT 2026 REAL-TSE Challenge: Real-world Target Speaker Extraction from Conversational Recordings

Shuai Wang¹, Zihan Qian¹, Ke Zhang², Jiangyu Han³, Zikai Liu⁴, Xiaoyang Yu¹, Haoyu Li¹

Marc Delcroix⁵, Kai Yu⁶, Lei Xie⁴, Ming Li² and Haizhou Li²

¹Nanjing University, China, ²Chinese University of Hong Kong (Shenzhen), China

³Brno University of Technology, Czechia, ⁴Northwestern Polytechnical University, China

⁵NTT, Inc., Japan, ⁶Shanghai Jiao Tong University, China

realtse.challenge@gmail.com

Abstract—We introduce the REAL-TSE Challenge¹, an IEEE SLT 2026 satellite challenge on target speaker extraction (TSE) from real conversational recordings. Given a multi-speaker mixture and one or more enrollment utterances from a target speaker, participating systems must recover only the target speech. Unlike simulated read-speech benchmarks, REAL-TSE evaluates Mandarin and English recordings that contain natural overlap, reverberation, noise, channel mismatch, and conversational dynamics. The challenge defines two complementary tracks: an *Online* track for low-latency streaming extraction and an *Offline* track for full-context processing. Systems are evaluated with Token Error Rate (TER), Speaker Similarity (SpkSim), DNSMOS, and target-speaker activity F1. This overview paper describes the task definition, datasets, baselines, evaluation protocol, submitted systems, condition-wise findings, and lessons for future real-world TSE benchmarks.

I. INTRODUCTION

Target Speaker Extraction (TSE) aims to isolate a specified speaker from a multi-talker recording using auxiliary cues such as enrollment utterances [1]–[3]. Robust TSE is an enabling technology for teleconferencing, hearing assistance, smart assistants, and speaker-aware meeting transcription.

Despite substantial progress, most widely used TSE evaluations still rely on simulated mixtures such as LibriMix [4] and WSJ0-2Mix [5], where clean read-speech utterances are summed at predefined signal-to-noise ratios. These benchmarks are reproducible and analytically convenient, but they do not fully capture the conditions that make conversational extraction difficult: uncontrolled loudness variation, real reverberation, ambient noise, turn-taking, reactive overlap, disfluencies, and non-verbal vocalizations. Existing challenges have advanced far-field ASR and speech enhancement [6]–[10], yet they do not directly benchmark real-recorded TSE under both low-latency and full-context settings.

The **REAL-TSE Challenge** addresses this gap with four design choices. First, it evaluates real Mandarin and English conversational recordings, combining data derived from REAL-T [11] with an independently collected unseen subset. Second, it separates online and offline tracks so that streaming and

batch systems can be assessed under appropriate constraints. Third, it uses a multi-dimensional scoring protocol that covers intelligibility, speaker consistency, perceptual quality, and target-speaker activity. Fourth, it releases baselines and validation tools built on WESEP [12]² together with an official toolkit for submission checking and metric computation³. The challenge collected 24 systems from 18 teams, whose results are available on the Leaderboard.⁴

The rest of this paper is organized as follows. Section II describes the tracks and metrics. Section III summarizes the data. Section IV presents the baselines. Section V provides an overview of the submitted systems. Section VI analyzes EVAL-2 condition-wise results. Section VIII discusses overall lessons, and Section IX concludes.

II. CHALLENGE SETTINGS

Each REAL-TSE trial consists of a real multi-speaker mixture x and an enrollment utterance e from the target speaker. The system output is an estimate $\hat{s}$ containing the target speech while suppressing non-target speakers and non-speech artifacts.

A. Track 1: Online Target Speaker Extraction

The online track targets latency-sensitive applications such as hearing devices and real-time communication. Systems must operate frame-wise or chunk-wise without full-utterance buffering, and their end-to-end algorithmic latency must not exceed 100 ms. The latency budget includes future context introduced by STFT/iSTFT, overlap-add, look-ahead frames, smoothing, post-processing, and chunk accumulation, but excludes hardware computation and I/O time. Effective latency is verified with a perturbation-based response-delay test.

B. Track 2: Offline Target Speaker Extraction

The offline track targets batch scenarios such as meeting transcription and audio post-production. Systems may use the full utterance, global context, arbitrary model classes, and

¹<https://real-tse.github.io/challenge/>

²<https://github.com/REAL-TSE/wesep-real-tse>

³<https://github.com/REAL-TSE/REAL-TSE-Challenge>

⁴<https://real-tse.github.io/challenge/#rankings>

unrestricted inference cost. However, each mixture–enrollment pair must still be processed independently, without metadata or information from other evaluation samples.

C. Evaluation Metrics

Systems are ranked by dense ranking on four metrics, and the final rank is the average of the four metric ranks. The metrics intentionally capture complementary aspects of TSE quality. As discussed in Sections V and VIII, this multi-objective design enhances the comprehensiveness of evaluation but also highlights the challenges associated with metric-aware over-optimization.

a) *Intelligibility—Token Error Rate (TER)*: TER measures ASR transcription accuracy, using word error rate for English and character error rate for Mandarin. Lower TER indicates better preservation of the target speaker’s linguistic content. The official ASR backbone is Zipformer [13].

b) *Speaker Consistency—Speaker Similarity (SpkSim)*: SpkSim is the cosine similarity between speaker embeddings of $\hat{s}$ and e , computed with the WeSpeaker ResNet-34 encoder [14]. Higher scores indicate better target-speaker consistency. Tables and figures abbreviate this metric as SIM.

c) *Perceptual Quality—DNSMOS*: DNSMOS is a non-intrusive perceptual-quality estimator reflecting artifacts, distortion, and residual noise. The challenge initially considered the DNSMOS OVRL [15] sub-score for perceptual-quality ranking. However, post-submission reliability analysis and subjective listening tests showed that OVRL suffered from metric-specific over-optimization in this competitive setting. We therefore **adopt DNSMOS-P808 [16] as the official perceptual-quality metric**, which appeared not to have been the target of over-optimization, and report OVRL only for reference. Details are provided in Section VII.

d) *Target-Speaker Activity—F1 Score*: F1 measures whether speech is emitted when, and only when, the target speaker is active. It combines temporal precision and recall over activity regions detected with FireRedVAD [17].

III. DATASETS

Most TSE benchmarks are simulated from clean read speech [18], which simplifies evaluation but introduces a mismatch with real reverberation, loudness variation, noise, and conversational turn-taking. REAL-TSE therefore grounds all evaluation data in real conversational recordings where overlap, speaker behavior, and recording-channel effects occur naturally.

A. Challenge Splits and Statistics

Table I summarizes the three challenge splits. Each sample is a released mixture–enrollment pair, and multiple samples may share mixture or enrollment files. In total, REAL-TSE contains 6,991 trials over 2,309 mixtures and 11.3 h of mixture audio in Mandarin and English.

Table II reports statistics for mixture duration, enrollment duration, overlap ratio, and target-speaker activity ratio.

TABLE I
STATISTICS OF THE CHALLENGE DATASET. SAMPLES ARE MIXTURE-ENROLLMENT PAIRS; MIXTURES AND ENROLLMENTS ARE UNIQUE WAV COUNTS; TARGET SPEAKERS ARE DISTINCT SPEAKERS.

	DEV	EVAL-1	EVAL-2
Samples	1,991	2,000	3,000
Mixtures	528	595	1,186
Enrollments	242	183	1,919
Target speakers	64	49	77
Mixture audio (h)	2.60	2.98	5.67
Enrollment audio (h)	0.67	0.55	4.05

TABLE II
MEAN VALUES OF THE MAIN CHARACTERISTICS OF EACH DATA SPLIT. THE NUMBERS IN PARENTHESES SHOW THE MIN AND MAX VALUES.

Quantity	DEV	EVAL-1	EVAL-2
Mix dur. (s)	17.26 (6.24, 29.91)	18.23 (5.82, 29.99)	17.25 (5.48, 29.91)
Enroll dur. (s)	9.65 (5.00, 50.51)	10.94 (5.01, 59.82)	9.10 (3.00, 42.98)
Overlap ratio	0.49 (0.18, 0.99)	0.48 (0.18, 0.98)	0.53 (0.18, 0.95)
Target ratio	0.75 (0.27, 1.00)	0.75 (0.21, 1.00)	0.73 (0.31, 1.00)

B. Development Set

The development set contains 1,991 mixture–enrollment pairs from REAL-T [11], built from AISHELL-4 [10], Al-iMeeting [9], AMI [19], DipCo [20], and CHiME6 [7]. Each sample pairs a naturally overlapping mixture with a non-overlapping target-speaker enrollment segment of at least 5 seconds. The development set is intended for hyper-parameter tuning, comparison, and ablation, and must **not** be used for model training or fine-tuning.

C. Evaluation Set

The evaluation set contains **5,000 mixture–enrollment pairs** in two subsets (Table I).

a) *EVAL-1 (Seen, 2,000 pairs)*: EVAL-1 is drawn from the same source corpora as the development set but contains no overlapping samples. It measures performance under in-domain conditions.

b) *EVAL-2 (Unseen, 3,000 pairs)*: EVAL-2 is newly collected and covers meetings, cafés, homes, and in-vehicle conversations. Each conversation was synchronously recorded with high-quality microphones (H1/H2), a phone, and headset microphones. Target speakers also provided external single-speaker enrollment utterances. EVAL-2 tests generalization to unseen environments, recording devices, and enrollment–mixture channel conditions.

Table III summarizes EVAL-2 channel pairings, including matched- and cross-channel cases used later for channel-mismatch analysis. To ensure a fair blind evaluation, scenario labels, speaker identities, and device metadata are removed from the released evaluation samples; each pair must be processed *independently*.

D. Training Data Policy

REAL-TSE has no official training set. Participants may use open-source data and pretrained models, but they must document all data sources and checkpoints. Development/test

TABLE III

EVAL-2 MIXTURE×ENROLLMENT CHANNEL PAIRING (SAMPLE COUNTS).
H1/H2: HIGH-QUALITY MICROPHONES; PHONE: MOBILE-PHONE
RECORDING; HEADSET: SPEAKER-WORN CLOSE-TALK HEADSET;
EXTERNAL: ADDITIONAL SINGLE-SPEAKER ENROLLMENT RECORDINGS.

Mix.\Enr.	Enrollment channel					Total
	H1	H2	phone	headset	external	
H1	397	127	129	141	218	1,012
H2	142	415	128	145	179	1,009
phone	122	135	388	131	203	979
Total	661	677	645	417	600	3,000

TABLE IV
BASELINE MODEL VARIANTS.

Model	Conditioning method	Extractor
BSRNN_EMB	ECAPA-TDNN emb.	Non-causal
BSRNN_EMB_CAUSAL	ECAPA-TDNN emb.	Causal
BSRNN_TFMAP	TF-Map + Context	Non-causal
BSRNN_TFMAP_CAUSAL	TF-Map + Context	Causal

splits of AliMeeting, AISHELL-4, AMI, DipCo, and CHiME6 are forbidden for pre-training, training, or augmentation, whereas their official training splits are allowed. This open-training policy encourages diverse data and modeling strategies while exposing the practical upper bound of current approaches.

IV. BASELINE SYSTEMS

We provide four BSRNN-based baselines [21], [22] implemented with WESEP [12]. All baselines are trained on fully overlapped Libri2Mix-100 [4] at 16 kHz for 150 epochs. They compare two enrollment-conditioning methods under causal and non-causal extractors, as summarized in Table IV.

BSRNN_EMB uses a fixed speaker embedding from ECAPA-TDNN [14], [23], whereas BSRNN_TFMAP combines spectral TF-Map features with contextual embeddings [22].

The causal baselines keep the same conditioning modules but impose causality inside the BSRNN extractor. Specifically, they replace global normalization with feature-axis normalization and bidirectional temporal LSTMs with unidirectional LSTMs.

Table V reports the official baseline results. Because all baselines are trained only on synthetic English-only Libri2Mix mixtures, they should be interpreted as lower-bound references rather than optimized challenge systems.

V. OVERVIEW OF SUBMITTED SYSTEMS

Each track received valid submissions from 12 teams. A valid submission complied with the challenge rules, produced valid evaluation outputs, and was accompanied by a system report. Rather than describing each system individually, we distill the recurring design choices that most consistently affected performance. Three of them stand out: (i) how real conversational data is exploited through simulation and adaptation; (ii) how the extractor is designed and conditioned under (or without) the latency budget; and (iii) how training objectives, post-processing, and test-time behavior interact with the automatic metrics.

TABLE V

BASELINE RESULTS ON DEV, EVAL-1, AND EVAL-2. LOWER TOKEN ERROR RATE (TER) IS BETTER; HIGHER SIM, DNSMOS P808/OVRL, AND F1 ARE BETTER. GRAY NUMBERS DENOTE THE PREVIOUSLY REPORTED DNSMOS OVRL SCORES.

Set	Model	TER	SIM	DNSMOS		F1
				P808	OVRL	
DEV	BSRNN_EMB	0.693	0.501	2.82	1.66	0.841
	BSRNN_EMB_CAUSAL	0.705	0.492	2.69	1.63	0.829
	BSRNN_TFMAP	0.703	0.521	2.72	1.50	0.838
	BSRNN_TFMAP_CAUSAL	0.652	0.535	2.76	1.56	0.844
EVAL-1	BSRNN_EMB	0.816	0.507	2.91	1.91	0.827
	BSRNN_EMB_CAUSAL	0.806	0.500	2.76	1.78	0.807
	BSRNN_TFMAP	0.837	0.535	2.80	1.75	0.822
	BSRNN_TFMAP_CAUSAL	0.801	0.553	2.84	1.79	0.837
EVAL-2	BSRNN_EMB	0.838	0.357	2.85	1.80	0.830
	BSRNN_EMB_CAUSAL	0.811	0.370	2.72	1.67	0.831
	BSRNN_TFMAP	0.839	0.382	2.73	1.52	0.833
	BSRNN_TFMAP_CAUSAL	0.808	0.391	2.75	1.59	0.835

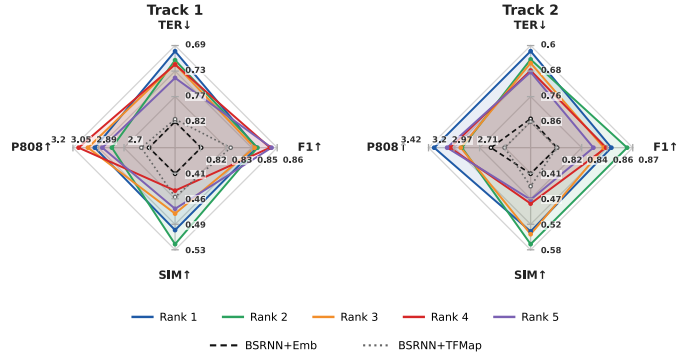


Fig. 1. Metric balance of the top-5 valid systems in each track.

Fig. 1 compares the top-five systems in each track with two baseline systems across the four official metrics. All metrics are normalized to a common scale, with TER reversed so that higher values consistently indicate better performance. Top systems substantially outperform both baselines across nearly all metrics. Nevertheless, **no leading system uniformly dominates across all evaluation dimensions**. Instead, the systems occupy distinct regions of the metric space: some perform better in speech intelligibility or target-speaker activity detection, whereas others more effectively preserve speaker identity or perceptual quality. These complementary strengths highlight the challenging multi-objective nature of real-world TSE, which requires a system to simultaneously preserve linguistic content, accurately detect target-speaker activity, maintain speaker identity, and produce perceptually natural speech.

A. Data Simulation and Real-Data Adaptation

Most teams synthesized mixtures from publicly available speech corpora such as LibriSpeech [24]/LibriMix [4], Vox-Celeb [25], CN-Celeb [26], AISHELL [10], [27], VCTK [28], and EARS [29]; added noise from WHAM! [30], MUSAN [31], DEMAND [32], or DNS Challenge data [33]; and introduced

reverberation with measured [34]/simulated RIRs [35], while varying the number of speakers, SNR/SIR, RT60, chunk length, and enrollment quality. Many teams also injected target-absent or noise-only segments, forcing systems not only to extract the target when present but also to suppress non-target speech and avoid false extraction.

The more consequential trend was **adaptation to real conversational data**. Because real-world mixtures rarely come with clean target references, competitive systems generated pseudo-labels using guided source separation, offline teacher models, model averaging, or self-training. The strongest approaches repeated this process over multiple rounds, gradually progressing from fully overlapped synthetic data to meeting-like mixtures and, ultimately, real far-field recordings. Across the reported results, real-data adaptation consistently outperformed training based solely on synthetic data. Notably, the top three teams either incorporated real recordings into their training pipelines or relied exclusively on real data during training.

B. Extractor Design and Speaker Conditioning

The latency constraint sharply narrowed the Track 1 design space: nearly all online systems were compact, causal discriminative extractors (mainly BSRNN and TF-GridNet [36] variants) adapted with causal normalization, unidirectional recurrence, and controlled look-ahead. Track 2 explored a much broader space, including SSL-based attractor initialization, diarization-aware front ends, and generative refinement based on diffusion, flow matching, or vocoder reconstruction. A recurring modeling question in both tracks was where and how to inject enrollment information, spanning global speaker embeddings, frame-level enrollment features, TF-Map conditioning, prefix-style tokens, and speaker-aware state modulation.

Crucially, **the results suggest substantial headroom beyond the current submitted systems**. The top-performing entries were all built on BSRNN-style backbones similar to the official baseline, and some used nearly the same extractor structure, yet achieved substantially better scores. This shows that realistic data simulation, real-data adaptation, pseudo-label generation and filtering, loss design, metric-aware training, and post-processing can dramatically boost an existing backbone. At the same time, this should not be interpreted as evidence that stronger backbones are unnecessary. Rather, it suggests that architecture and training recipe have not yet been fully co-optimized for real-world TSE: with similarly strong data preparation and adaptation pipelines, more expressive backbones such as TF-GridNet-style extractors may still have considerable room to improve.

C. Objectives, Post-Processing, and Metric-Aware Behavior

Reconstruction losses (SI-SDR/SI-SNR, multi-resolution STFT) were routinely combined with auxiliary terms aligned to the official metrics, including speaker-similarity, DNSMOS-related, ASR/token-level, and VAD/activity losses, marking a clear shift from single-objective separation toward multi-objective optimization. Post-processing followed the same logic, particularly in Track 2: several teams appended enhancement

or vocoder-based reconstruction modules, or produced raw and enhanced branches and selected between them using speaker-similarity criteria. Such post-processing typically improved DNSMOS but could reduce SpkSim or alter target activity, and should be viewed as a trade-off rather than a uniformly beneficial step.

This metric-aware behavior, however, exposed a fundamental weakness of the evaluation itself. Optimizing a neural metric can produce artifacts that raise the score without improving perceived quality. The system reports revealed borderline practices such as branch selection based on automatic scores and repeated use of the official validation scripts during development; in one extreme case, *adversarial waveform perturbations* were applied to inflate SpkSim and DNSMOS without genuinely improving extraction. These cases motivate more precise rules for future challenge **that forbid the use of non-intrusive metrics in the final evaluation** during development (see Section VIII for details).

For Track 1, algorithmic latency mainly came from STFT/iSTFT delay, hop buffering, look-ahead, chunk accumulation, and other forms of non-causal context. Most submitted systems reported latency in the 20–100 ms range. The perturbation-based latency script was important for verifying these claims, since analytical estimates and measured delays did not always agree. In some cases, systems reported substantial look-ahead but showed relatively small measured delay; in others, latency was underestimated because buffering, normalization, or other implementation details were not fully accounted for. These observations suggest that future online TSE benchmarks should require both a transparent breakdown of analytical latency and a standardized measurement procedure.

VI. CONDITION-WISE ANALYSIS OF EVAL-2 RESULTS

Beyond system-report summaries, REAL-TSE supports aggregate analysis of how performance changes under real-world recording conditions. We focus on EVAL-2 because, unlike DEV and EVAL-1, it was newly recorded by the organizers to cover diverse scenarios, devices, and enrollment-mixture channel pairings. We will analyze aggregate trends by recording device/ channel pairing, target-speaker activity ratio, and scenario.

A. Device and Channel Mismatch Effects

We begin by examining channel effects between enrollment and mixture recordings. Figure 3 shows the performance measure by the four metrics with different microphones for the mixture/enrollment pairs. The results are the average over all valid submitted systems. As shown in Fig. 3, **the mixture recording device has a stronger effect than the enrollment device**. Mixtures recorded by the H2 microphone achieved consistently lower performance than H1 and Phone. This matches the recording setup, in which H2 is the farthest microphone and captures more reverberation and environmental interference. Robustness to far-field mixture channels thus remains a key open challenge for real-world TSE. Interestingly, the mismatch between the mixture and the enrollment recording

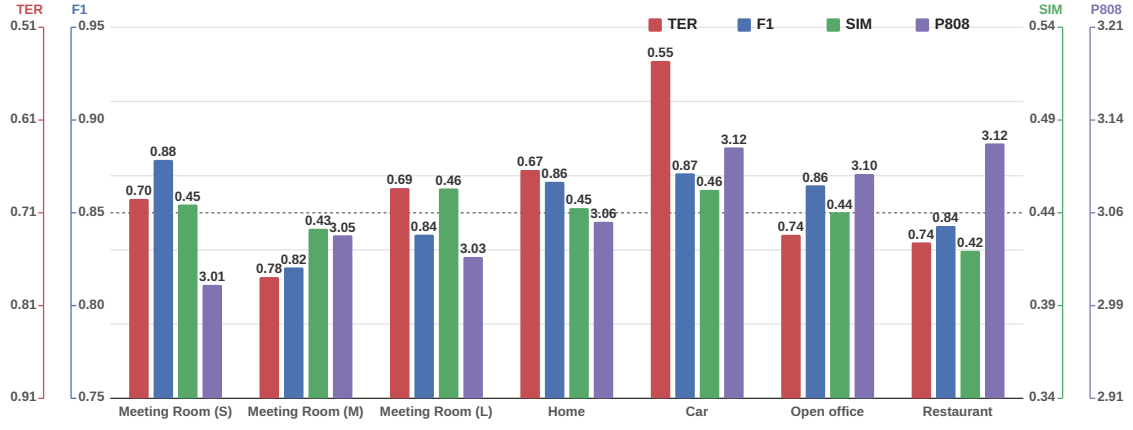


Fig. 2. Scenario-wise performance on the EVAL-2 set, computed by averaging all valid submissions separately for CN and EN and then taking a CN/EN macro average over shared scenarios.

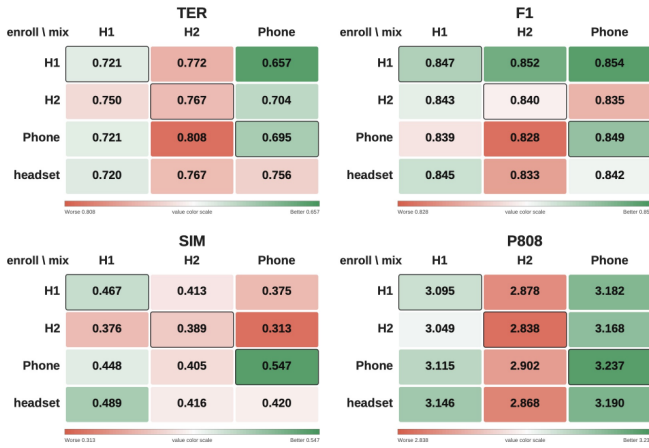


Fig. 3. Enrollment-device by mixture-device performance matrix on EVAL-2.

device had different impacts depending on the metrics, but overall, the systems appear not over-sensitive to such a mismatch, except for speaker similarity. This may be due to the metric computation, which uses enrollment as a reference and may thus be sensitive to channel mismatch.

B. Target Ratio and Utterance-level Activity

We further examine the relationship between utterance-level difficulty and the target ratio in Fig. 4, measured as the proportion of target-speaker activity in the mixture.

Overall, target ratio shows the clearest effect on target-activity F1. As the target speaker occupies a larger portion of the mixture, F1 increases, indicating more reliable detection and preservation of target-active regions. TER shows a weaker decreasing trend, suggesting that target ratio alone does not fully explain utterance-level intelligibility errors.

C. Scenario and Conversational Difficulty

We next analyze scenario-level variation by grouping EVAL-2 samples according to the recording scenario and computing

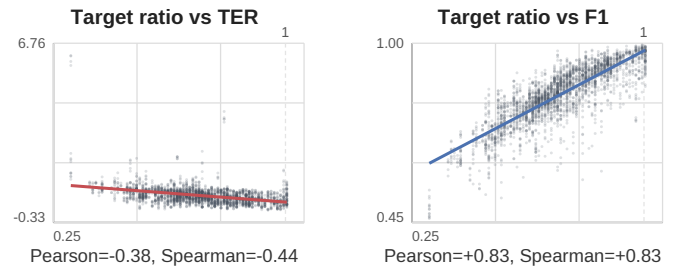


Fig. 4. Relationship between target ratio and F1/TER on EVAL-2.

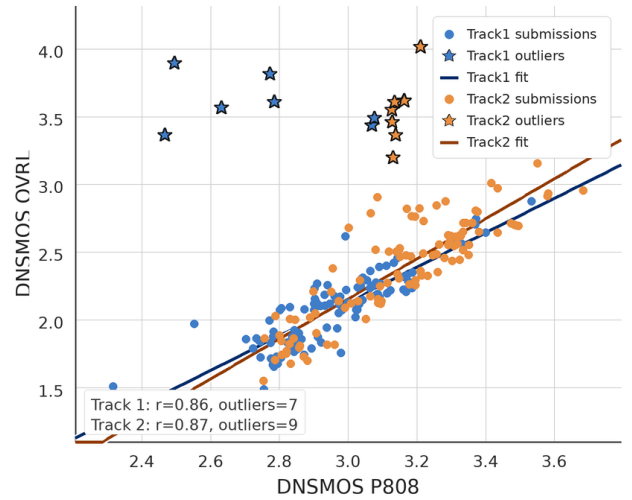


Fig. 5. Relationship between DNSMOS OVRL and DNSMOS P808 on participant submissions. The high-OVRL outliers do not show corresponding gains in P808, indicating metric-specific over-optimization rather than a consistent improvement in perceptual quality.

average performance for each official metric. The goal is to examine whether coarse scene labels, such as meeting rooms, restaurants, homes, and in-vehicle conversations, reliably predict TSE difficulty.

TABLE VI
CORRELATION BETWEEN HUMAN MOS AND DNSMOS VARIANTS FOR
TOP-FIVE FINAL SUBMISSIONS.

Scope	n	DNSMOS-OVRL		DNSMOS-P808		DNSMOS-PRO [39]	
		LCC	SRCC	LCC	SRCC	LCC	SRCC
Track 1	500	+0.003	+0.040	+0.453	+0.380	+0.156	+0.115
Track 2	600	+0.182	+0.186	+0.466	+0.463	+0.229	+0.234
Overall	1100	+0.165	+0.186	+0.489	+0.460	+0.233	+0.230

The observed trends do not consistently follow intuitive acoustic expectations. Meeting-room recordings might appear easier, and in-vehicle recordings seem manageable because the car interior is small and the recording geometry is relatively fixed. Conversely, restaurant-like scenarios might be expected to be more difficult because of stronger background noise and less controlled conversational behavior. However, the aggregate metric patterns in Fig. 2 do not follow this ordering consistently. The nominal scenario label alone is therefore insufficient to determine real-world TSE difficulty.

A likely reason is that EVAL-2 conversations were deliberately recorded in a natural and topic-driven manner. Speakers interacted freely rather than following a tightly controlled script, so each scenario contains substantial variability in turn-taking, overlap, speaker distance, loudness, device placement, and local noise events. Scenario-wise statistics should therefore be interpreted as a coarse diagnostic view rather than a strict ranking of acoustic difficulty.

VII. REVISITING DNSMOS RELIABILITY

Near the final stage of the challenge, we observed anomalous behavior in DNSMOS OVRL, the perceptual-quality score originally considered for ranking. Some submissions achieved unusually high OVRL scores that did not align with listening impressions or with other DNSMOS variants. We therefore conducted a post-submission reliability analysis using the submissions from each track.⁵

Fig. 5 shows the main symptom: several systems obtain high OVRL scores without comparable gains in P808. Since both scores are intended to estimate perceptual quality, such outliers suggest metric-specific over-optimization rather than uniform improvement in extracted-speech quality. To quantify this effect, Table VI compares DNSMOS variants against human MOS using linear correlation (LCC [37]) and rank correlation (SRCC [38]). OVRL is weakly correlated with human MOS, especially in Track 1, whereas P808 shows consistently higher agreement across both tracks. Based on this evidence, we use DNSMOS-P808 as the official perceptual-quality metric and keep OVRL only as a reference score.

Together, these observations highlight a broader limitation of reference-free neural evaluation metrics. When such metrics are public and can be used directly during system development,

⁵A similar issue was also spotted for the speaker similarity, but it appeared to have less of an impact for evaluation. Due to space constraints, we focus thus our discussion on DNSMOS.

they may be unintentionally over-optimized or even exploited by model training, post-processing, or selection strategies. As a result, the metric can lose its meaning, even if systems achieve higher automatic scores.

VIII. OVERALL LESSONS

The submissions suggest three lessons for future real-world TSE evaluations. First, most teams focused less on proposing fundamentally new extractor architectures and more on making existing backbones work under realistic conditions. Strong systems commonly combined BSRNN- or TF-GridNet-style extractors with careful data simulation, real-data adaptation, pseudo-label filtering, multi-objective training, post-processing, and latency control. This does not imply that architecture is unimportant; rather, it shows that backbone design and training recipes need to be co-optimized, and that stronger models may still have substantial headroom when paired with equally strong data preparation.

Second, REAL-TSE confirms that TSE quality is inherently multi-dimensional. No system consistently dominates TER, SpkSim, DNSMOS P808, and target-activity F1 at the same time. Different applications may therefore prefer different operating points: meeting transcription may emphasize intelligibility, hearing-assistance and communication scenarios may prioritize latency and perceptual quality, and speaker-aware applications may require stronger identity preservation and activity accuracy. Future leaderboards should report both aggregate rankings and use-case-specific views.

Third, metric reliability and real-world difficulty must be treated as part of benchmark design. The DNSMOS analysis shows that public reference-free metrics can be over-optimized, so future challenges should require transparent reporting of metric usage, output selection, post-processing, and latency measurement. Meanwhile, EVAL-2 shows that difficulty cannot be explained by coarse scenario labels alone: target-speaker activity, channel mismatch, overlap behavior, speaker distance, device placement, and conversation dynamics all affect performance. Future real-recorded TSE benchmarks should therefore characterize difficulty at the sample, interaction, and channel level, not only at the nominal scenario level.

IX. CONCLUSIONS

The REAL-TSE Challenge takes a step toward bridging controlled laboratory benchmarks and real-world target speaker extraction. By grounding evaluation in authentic Mandarin and English conversational recordings, and by providing both online and offline tracks, the challenge enables a multi-faceted assessment of current TSE systems. The open-training policy and multi-dimensional scoring framework encourage creative and practically deployable solutions, while the submitted systems show that progress depends jointly on data construction, speaker-conditioning design, multi-objective optimization, post-processing, and transparent latency reporting. We hope REAL-TSE will help move the community toward TSE systems that are robust, generalizable, and directly applicable to real conversational scenarios.

REFERENCES

- [1] Q. Wang, H. Muckenhirn, K. Wilson, P. Sridhar, Z. Wu, J. Hershey, R. A. Saurous, R. J. Weiss, Y. Jia, and I. L. Moreno, "VoiceFilter: Targeted voice separation by speaker-conditioned spectrogram masking," in *Proc. Interspeech*, 2019, pp. 2728–2732.
- [2] C. Xu, W. Rao, E. S. Chng, and H. Li, "SpEx: Multi-scale time domain speaker extraction network," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 1370–1384, 2020.
- [3] K. Zmolikova, M. Delcroix, T. Ochiai, K. Kinoshita, J. Černocký, and D. Yu, "Neural target speech extraction: An overview," *IEEE Signal Processing Magazine*, vol. 40, no. 3, pp. 8–29, 2023.
- [4] J. Cosentino, M. Pariente, S. Cornell, A. Deleforge, and E. Vincent, "LibriMix: An open-source dataset for generalizable speech separation," *arXiv preprint arXiv:2005.11262*, 2020.
- [5] J. R. Hershey, Z. Chen, J. L. Roux, and S. Watanabe, "Deep clustering: Discriminative embeddings for segmentation and separation," in *Proc. ICASSP*, 2016, pp. 31–35.
- [6] J. Barker, S. Watanabe, E. Vincent, and J. Trmal, "The Fifth 'CHiME' Speech Separation and Recognition Challenge: Dataset, Task and Baselines," in *Proc. Interspeech*, 2018, pp. 1561–1565.
- [7] S. Watanabe, M. Mandel, J. Barker, E. Vincent, A. Arora, X. Chang, S. Khudanpur, V. Manohar, D. Povey, D. Raj, D. Snyder, A. S. Subramanian, J. Trmal, B. B. Yair, C. Boeddeker, Z. Ni, Y. Fujita, S. Horiguchi, N. Kanda, T. Yoshioka, and N. Ryant, "CHiME-6 Challenge: Tackling Multispeaker Speech Recognition for Unsegmented Recordings," in *6th International Workshop on Speech Processing in Everyday Environments (CHiME 2020)*, 2020, pp. 1–7.
- [8] R. Sutherland, J. Clarke, H. Elghazaly, T. Kuebert, M. Lugger, S. Petrausch, J. A. Ortiz, B. Xu, S. Goetze, and J. Barker, "Descriptor: Enhancing conversations for the hearing impaired in the 9th computational hearing in multisource environments challenge (CHiME9 ECHI)," *IEEE Data Descriptions*, vol. 3, pp. 73–81, 2026.
- [9] F. Yu, S. Zhang, Y. Fu, L. Xie, S. Zheng, Z. Du, W. Huang, P. Guo, Z. Yan, B. Ma, X. Xu, and H. Bu, "M2MeT: The ICASSP 2022 multi-channel multi-party meeting transcription challenge," in *Proc. ICASSP*. IEEE, 2022, pp. 6167–6171.
- [10] Y. Fu, L. Cheng, S. Lv, Y. Jv, Y. Kong, Z. Chen, Y. Hu, L. Xie, J. Wu, H. Bu, X. Xu, J. Du, and J. Chen, "AISHELL-4: An open source dataset for speech enhancement, separation, recognition and speaker diarization in conference scenario," in *Proc. Interspeech*, 2021, pp. 3665–3669.
- [11] S. Li, S. Wang, J. Han, K. Zhang, W. Wang, and H. Li, "REAL-T: Real Conversational Mixtures for Target Speaker Extraction," in *Proc. Interspeech*, 2025, pp. 1923–1927.
- [12] S. Wang, K. Zhang, S. Lin, J. Li, X. Wang, M. Ge, J. Yu, Y. Qian, and H. Li, "WeSep: A Scalable and Flexible Toolkit Towards Generalizable Target Speaker Extraction," in *Proc. Interspeech*, 2024, pp. 4273–4277.
- [13] Z. Yao, L. Guo, X. Yang, W. Kang, F. Kuang, Y. Yang, Z. Jin, L. Lin, and D. Povey, "Zipformer: A faster and better encoder for automatic speech recognition," in *Proc. ICLR*, vol. 2024, 2024, pp. 44 440–44 455.
- [14] H. Wang, C. Liang, S. Wang, Z. Chen, B. Zhang, X. Xiang, Y. Deng, and Y. Qian, "Wespeaker: A research and production oriented speaker embedding learning toolkit," in *Proc. ICASSP*. IEEE, 2023, pp. 1–5.
- [15] C. K. A. Reddy, V. Gopal, and R. Cutler, "DNSMOS P.835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *Proc. ICASSP*, 2022, pp. 886–890.
- [16] C. K. Reddy, V. Gopal, and R. Cutler, "DNSMOS: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *Proc. ICASSP*. IEEE, 2021, pp. 6493–6497.
- [17] K. Xu, Y. Jia, K. Huang, J. Chen, W. Li, K. Liu, F.-L. Xie, X. Tang, and Y. Hu, "Firedaser2s: A state-of-the-art industrial-grade all-in-one automatic speech recognition system," *arXiv preprint arXiv:2603.10420*, 2026.
- [18] J. B. Allen and D. A. Berkley, "Image method for efficiently simulating small-room acoustics," *The Journal of the Acoustical Society of America*, vol. 65, no. 4, pp. 943–950, 1979.
- [19] J. Carletta, S. Ashby, S. Bourban, M. Flynn, M. Guillemot, T. Hain, J. Kadlec, V. Karaiskos, W. Kraaij, M. Kronenthal, G. Lathoud, M. Lincoln, A. Lisowska, I. McCowan, W. Post, D. Reidsma, and P. Wellner, "The AMI meeting corpus: A pre-announcement," in *Machine Learning for Multimodal Interaction (MLMI 2005)*, ser. Lecture Notes in Computer Science, S. Renals and S. Bengio, Eds., vol. 3869. Berlin, Heidelberg: Springer, 2006, pp. 28–39.
- [20] M. Van Segbroeck, A. Zaid, K. Kutsenko, C. Huerta, T. Nguyen, X. Luo, B. Hoffmeister, J. Trmal, M. Omologo, and R. Maas, "DiPCo – Dinner Party Corpus," in *Proc. Interspeech*, 2020, pp. 434–436.
- [21] Y. Luo and J. Yu, "Music source separation with band-split RNN," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 1215–1225, 2023.
- [22] K. Zhang, J. Li, S. Wang, Y. Wei, Y. Wang, Y. Wang, and H. Li, "Multi-level speaker representation for target speaker extraction," in *Proc. ICASSP*, 2025, pp. 1–5.
- [23] B. Desplanques, J. Thienpondt, and K. Demuynck, "ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification," in *Proc. Interspeech*, 2020, pp. 3830–3834.
- [24] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: an asr corpus based on public domain audio books," in *Proc. ICASSP*. IEEE, 2015, pp. 5206–5210.
- [25] A. Nagrani, J. S. Chung, and A. Zisserman, "Voxceleb: A large-scale speaker identification dataset," in *Proc. Interspeech*, 2017, pp. 2616–2620.
- [26] Y. Fan, J. Kang, L. Li, K. Li, H. Chen, S. Cheng, P. Zhang, Z. Zhou, Y. Cai, and D. Wang, "CN-Celeb: A challenging chinese speaker recognition dataset," in *Proc. ICASSP*, 2020, pp. 7604–7608.
- [27] H. Bu, J. Du, X. Na, B. Wu, and H. Zheng, "Aishell-1: An open-source mandarin speech corpus and a speech recognition baseline," in *2017 20th Conference of the Oriental Chapter of the International Coordinating Committee on Speech Databases and Speech I/O Systems and Assessment (O-COCOSDA)*, 2017, pp. 1–5.
- [28] J. Yamagishi, C. Veaux, and K. MacDonald, "CSTR VCTK Corpus: English multi-speaker corpus for CSTR voice cloning toolkit," 2019, sound dataset. [Online]. Available: <https://doi.org/10.7488/ds/2645>
- [29] J. Richter, Y.-C. Wu, S. Krenn, S. Welker, B. Lay, S. Watanabe, A. Richard, and T. Gerkmann, "EARS: An Anechoic Fullband Speech Dataset Benchmarked for Speech Enhancement and Dereverberation," in *Proc. Interspeech*, 2024, pp. 4873–4877.
- [30] G. Wichern, J. Antognini, M. Flynn, L. R. Zhu, E. McQuinn, D. Crow, E. Manilow, and J. L. Roux, "WHAM!: Extending Speech Separation to Noisy Environments," in *Interspeech 2019*, 2019, pp. 1368–1372.
- [31] D. Snyder, G. Chen, and D. Povey, "Musan: A music, speech, and noise corpus," 2015. [Online]. Available: <https://arxiv.org/abs/1510.08484>
- [32] J. Thiemann, N. Ito, and E. Vincent, "DEMAND: A collection of multi-channel recordings of acoustic noise in diverse environments," 2013, version 1.0. [Online]. Available: <https://doi.org/10.5281/zenodo.1227121>
- [33] C. K. Reddy, V. Gopal, R. Cutler, E. Beyrami, R. Cheng, H. Dubey, S. Matusevych, R. Aichner, A. Aazami, S. Braun, P. Rana, S. Srinivasan, and J. Gehrke, "The INTERSPEECH 2020 Deep Noise Suppression Challenge: Datasets, Subjective Testing Framework, and Challenge Results," in *Proc. Interspeech*, 2020, pp. 2492–2496.
- [34] I. Szöke, M. Skácel, L. Mošner, J. Paliesek, and J. Černocký, "Building and evaluation of a real room impulse response dataset," *IEEE Journal of Selected Topics in Signal Processing*, vol. 13, no. 4, pp. 863–876, 2019.
- [35] Y. Luo and R. Gu, "Fast random approximation of multi-channel room impulse response," in *2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, 2024, pp. 449–454.
- [36] Z.-Q. Wang, S. Cornell, S. Choi, Y. Lee, B.-Y. Kim, and S. Watanabe, "TF-GridNet: Integrating full-and sub-band modeling for speech separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 3221–3236, 2023.
- [37] K. Pearson, "Notes on the history of correlation," *Biometrika*, vol. 13, no. 1, pp. 25–45, 1920.
- [38] C. Spearman, "The proof and measurement of association between two things," 1961.
- [39] F. Cumlin, X. Liang, V. Ungureanu, C. K. Reddy, C. Schüldt, and S. Chatterjee, "DNSMOS Pro: A reduced-size DNN for probabilistic MOS of speech," in *Proc. Interspeech*, 2024, pp. 4818–4822.